USN | | | | | | | | | | [23TCSE312]

THIRD SEMESTER B.Tech EXAMINATION, NOVEMBER-2024

CLOUD SECURITY

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	What are the primary components of cloud infrastructure?	**L2**	**CO1**	**02**
	b.	List out the types of hypervisors.	**L2**	**CO1**	**02**
	c.	What is a virtual private cloud (VPC), and how does it enhance security?	**L4**	**CO3**	**02**
	d.	How does the responsibility model change with SaaS compared to IaaS?	**L2**	**CO3**	**02**
	e.	What is a shared responsibility model in cloud security?	**L2**	**CO1**	**02**
		PART-B			**6 X 5=30**
2.	**a.**	Illustrate the main security concerns with hypervisors?	**L2**	**CO2**	**05**
		OR			
	b.	Discuss the best practices to secure a hypervisor?	**L2**	**CO3**	**05**
3.	**a.**	What are the main risks to consider in SaaS?	**L1**	**CO1**	**05**
		OR			
	b.	Describe hat is the role of multi-factor authentication (MFA) across IaaS, PaaS, and SaaS?	**L2**	**CO3**	**05**
4.	**a.**	Explain operational application security, and why is it important?	**L2**	**CO5**	**05**
		OR			
	b.	Describe the vulnerability management, and how does it support operational application security?	**L2**	**CO3**	**05**
5.	**a.**	What is encryption, and how does it differ from encoding and hashing? What are symmetric and asymmetric encryption, and where are they commonly used?	**L2**	**CO2**	**05**
		OR			
	b.	How do these concepts (encryption, key management, masking, tokenization) work together in a security strategy?	**L3**	**CO5**	**05**
6.	**a.**	Explain the importance of security testing, and why is it critical for software and cloud applications?	**L4**	**CO3**	**05**
		OR			

	b.	Illustrate the regulatory compliance implications of cloud adoption on Enterprise Risk Management.	L2	CO2	05
7.	a.	How can ERM frameworks adapt to cloud-specific risks through automation and advanced analytics?	L2	CO2	05
		OR			
	b.	Discuss in detail Risk Management Framework (RMF), and why is it essential for organizations?	L3	CO5	05
		PART-C		**6 X 10=60**	
8.	a.	What are the main security concerns in a virtualized environment, and how do they differ from traditional physical environments? What are the main security concerns in a virtualized environment, and how do they differ from traditional physical environments?	L1	C01	10
		OR			
	b.	Explain the Secure Boot, and how does it enhance the security of virtualized platforms? How does Trusted Platform Module (TPM) technology contribute to virtualization security?	L2	C01	10
9.	a.	Assess the key phases of the Cloud Secure Development Lifecycle (CSDLC), and what security practices are essential in each phase?	L4	C01	10
		OR			
	b.	Why is insufficient logging and monitoring a vulnerability in cloud applications, and what steps can organizations take to address this issue?	L3	C02	10
10.	a.	Design a cloud application architecture, and how does it differ from traditional on-premises architecture? How do cloud-native principles (e.g., microservices, containerization, and serverless) impact cloud application architecture?	L4	C03	10
		OR			
	b.	Explain the concepts of sandboxing, and why is it essential in application development and security? How does sandboxing work, and what are common sandboxing techniques?	L2	C01	10
11.	a.	What are the main types of security testing, and how do they differ? How does penetration testing in cloud environments differ from traditional environments?	L2	C02	10
		OR			

	b.	How has the adoption of cloud computing reshaped Enterprise Risk Management (ERM) frameworks? What are the major risks associated with cloud computing for enterprises, and how do they impact risk management strategies?	**L4**	**C02**	**10**
12.	**a.**	Describe a Service-Level Agreements (SLAs), and why are they critical in technology and cloud service contracts?	**L4**	**C02**	**10**
		OR			
	b.	Explain the key components of an SLA, and how do they contribute to effective service delivery?	**L1**	**C01**	**10**
13.	**a.**	Illustrate the essential elements of a cloud outsourcing contract, and why is each important?	**L4**	**C02**	**10**
		OR			
	b.	Why is data governance a critical consideration in cloud outsourcing contracts, and what are the key components of effective data governance clauses?	**L4**	**C02**	**10**